

# Curtis Fenner Tokyo, Japan

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I am a deeply curious software engineer, specialized in careful knowledge of programming language and database semantics and performance.

I have extensive professional experience designing, building, and maintaining backend systems in statically-typed languages, as well as experience developing frontend applications with functional frameworks like React.

## Experience

**Givery** Engineer, "Track Test" & "Common Systems" teams ..... Tokyo, JP. Jan 2023 – current

- » Track Test Team dev. lead (2025–): estimated, designed, and refined plans for long-term projects
- » Led architecture design, technology choice and modernization, and implementation of new product
  - » Introduced new Zod-based wrappers to improve type-safety & eliminate validation boilerplate
  - » Designed and implemented precise permission engine to make permissions & role checks standardized, flexible, and obviously correct
- » Redesigned slow SQL queries to eliminate load and change slow-loading pages to instantly loading ones
- » Refactored and documented internal scoring and submission logic to improve auditability, squash bugs, and increase extensibility

⚙ **Scala (Play), TypeScript (NestJS + Kysely)** with **Postgresql** (backends)  
⚙ **TypeScript (React)** (frontend)  
⚙ **AWS (ECS, Lambda, Rekognition, S3)**

**PayPay** Backend Engineer, "Online" team .... Remote, Jun '21 – Apr '22 .... Tokyo, JP. Apr 2022 – Jan 2023

- » Backend engineer on the Online team, which external APIs for accepting PayPay payments online
- » Designed, documented, and implemented new APIs used by merchants to accept PayPay payments
- » Designed and lead a new microservice, using requirements from Product Managers and other teams

⚙ **Kotlin & Java (Spring Boot)** with **MySQL** (backend)  
⚙ **AWS (EC2, KMS)**

**Square** Software Engineer, "Orders API" team ..... Atlanta, GA. Aug 2018 – Dec 2020

- » Backend engineer on the Orders API team, which operates both a public REST API and internal POS API
- » Refactored, tested, thoroughly documented, and simplified complex validation, transaction calculation, and public API versioning logic

⚙ **Java, MySQL, and Protobuf** (strongly consistent API server)  
⚙ **TypeScript (React)** (internal web UI)

## Education

**University of Michigan: Computer Science B.S.E, 4.0 GPA** ..... Ann Arbor, MI. 2014 – 2018

- » **Selected Coursework:** Distributed Systems (W2017), Grad. Programming Languages (F2017)
- » Teaching assistant for Distributed Systems using Golang (F2017)
- » World Finalist in 2017 ACM International Collegiate Programming Contest (ICPC)
- » Japanese Language I + II

I had 3 software engineering internships shipping real production code before graduating from university:

- » **Qualtrics** Seattle, WA. Summer 2017
  - ⚙ **Scala** and **Elasticsearch** (customer report generation)
    - » Redesigned statistical weighting formulas to fix error independently discovered
- » **Qumulo** Seattle, WA. Summer 2016
  - ⚙ **C** (filesystem server) and **Python** (test automation and code generation)
    - » Eliminated lock contention in a multithreaded cache to reduce file operation latency
- » **Square** San Francisco, CA. Summer 2015
  - ⚙ **Node.JS** and **Go** (microservice backends)
    - » Optimized and refactored public API server to halve average response time

**Japanese Language Proficiency Test N3 (JLPT/日本語能力試験)** ..... Tokyo – July 2024

## Selected Projects

**Tokyo Rail-Transit Isochrone** ..... [github.com/CurtisFenner/tokyo-transit-iso](https://github.com/CurtisFenner/tokyo-transit-iso)

- » Implemented routing algorithm and region geometry rendering algorithm on real transit schedule data
- ⚙ **TypeScript, MapLibre GL JS**

**Shiru Programming Language & Compiler** ..... [github.com/CurtisFenner/shiru-ts](https://github.com/CurtisFenner/shiru-ts)

- » Designed and implemented a toy programming language, compiler, and assertion verifier, including a PEG parser, a type-checker, a simple SMT solver, and a JavaScript code generator.
- ⚙ **TypeScript**